

Table 1 highlights the essential modules of Qt.

Name of module	Description
Qt Core	Contains all core non-graphical classes used by other modules
Qt GUI	Base classes for GUI components. It includes OpenGL
Qt Multimedia	Classes for audio, video, radio and camera functionality
Qt Multimedia Widgets	Widget-based classes for implementing multimedia functionality
Qt Network	Classes to make network programming easier and more portable
Qt QML	Classes for QML and JavaScript languages
Qt Quick	A declarative framework for building highly dynamic applications with custom user interfaces
Qt Quick Controls	Provides lightweight QML types for creating performant user interfaces for desktop, embedded, and mobile devices. They employ a simple styling architecture and are highly efficient
Qt Quick dialogues	These are for creating and interacting with system dialogues for Qt Quick application
Qt Quick Layouts	Layouts are used to arrange Qt Quick 2 based items in the user interface
Qt Quick Test	This is a unit test framework for QML applications, where the test cases are written as JavaScript functions
Qt SQL	Classes for database integration using SQL
Qt Test	Classes for unit testing Qt applications and libraries
Qt Widgets	Classes to extend Qt GUI with C++ widgets

- is a model/view architecture for item views and the QSql module, which also uses this architecture.
- *The Scribe Classes:* This is a framework for creating text documents, performing low-level text layout and writing OpenDocument files.
 - *Common Desktop Widgets:* These are optimised to fit to support every platform.
 - *Qt4 Main Window Classes:* These include Main Window, Toolbar, Menu and the docking architecture.
 - *Graphics View:* This framework provides a canvas for producing interactive graphics.
 - *QtNetwork Module:* This provides support for TCP, UDP and local sockets that are integrated with Qt's event model, including support for Secure Socket Layer (SSL) communications, network proxy servers and network bearer management.
 - Enhanced thread support allows signal-slot connections across threads and per-thread event loops.

- In addition, a framework for concurrent programming using Qt paradigms makes common threading tasks easier.
- *Resource system:* This is for embedding images and other resource files into executable files, and it makes it easier to deploy applications.
 - *Unit testing framework:* This is for Qt applications and libraries.

Qt modules

A module is used for a special purpose and there are even some add-on modules that are available on all supported platforms.

Top open source tools for Qt developers

In order to assist developers in Qt programming, the following top open source tools have been reviewed to help them locate and fix bugs, and solve difficult optimisation challenges.

1. GammaRay

GammaRay is a software introspection tool for Qt applications developed

by KDAB. GammaRay understands most of the core Qt components from a Qt perspective – QtQuick scene graphs, model/view structures, QTextDocuments, signal/slot activation, focus handling, GPU textures, QWidget, state machines and more – letting the developer observe and edit values at runtime in a natural way. GammaRay consists of two parts, the client and the probe. The client is the GUI that the user interacts with. The probe interacts very closely with the Qt application it analyses, using a number of internal APIs of Qt. This makes it necessary that the probe is built for exactly your Qt.

GammaRay allows the developers to work on a much higher level, with the same concepts as the frameworks they use. This is especially useful for the more complex Qt frameworks such as model/view, state machines or scene graphs. GammaRay addresses this by providing domain-specific debugging aids on a much higher level. It provides easy ways of navigating through the complex internal structures found